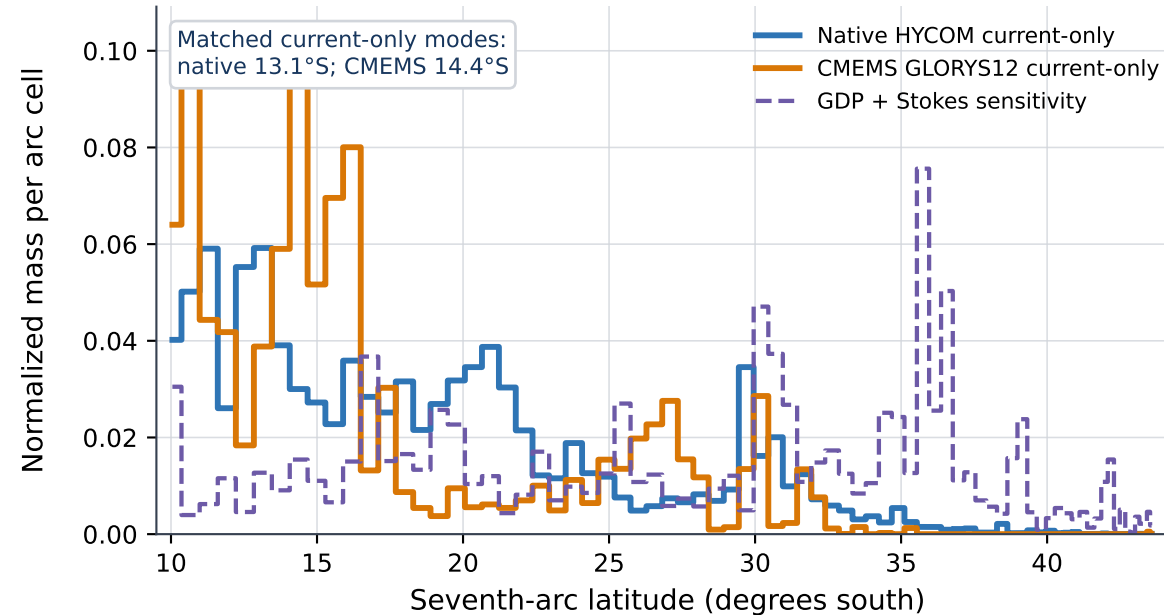


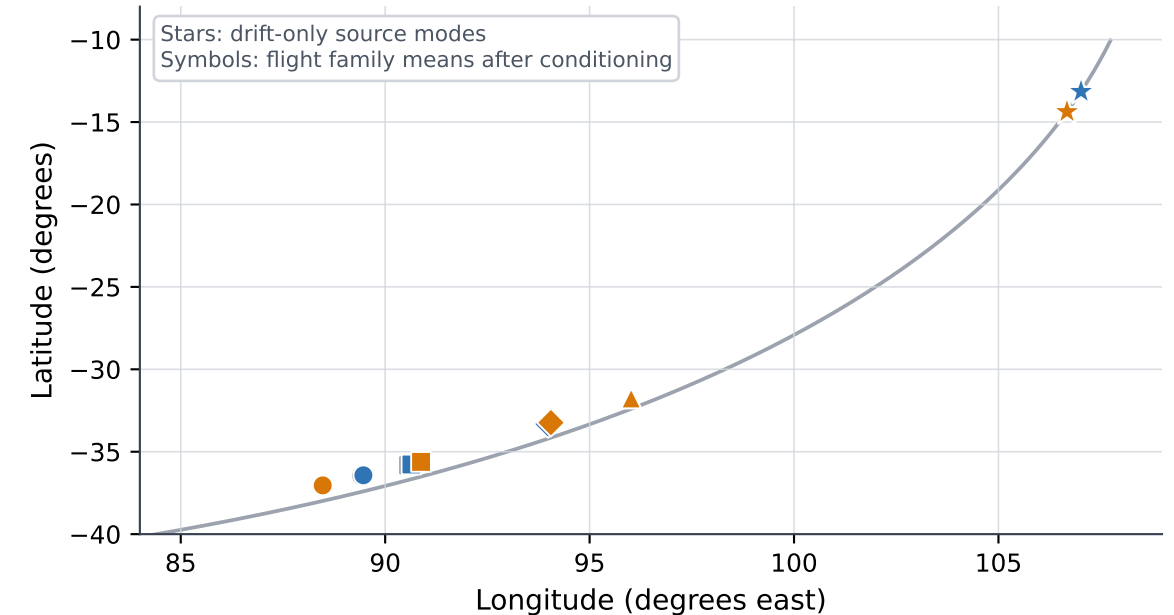
MH370 flaperon source and inferred crash-location diagnostics

Matched 8,192-particle currents-only source runs; four converged flight-control families; no family pooling

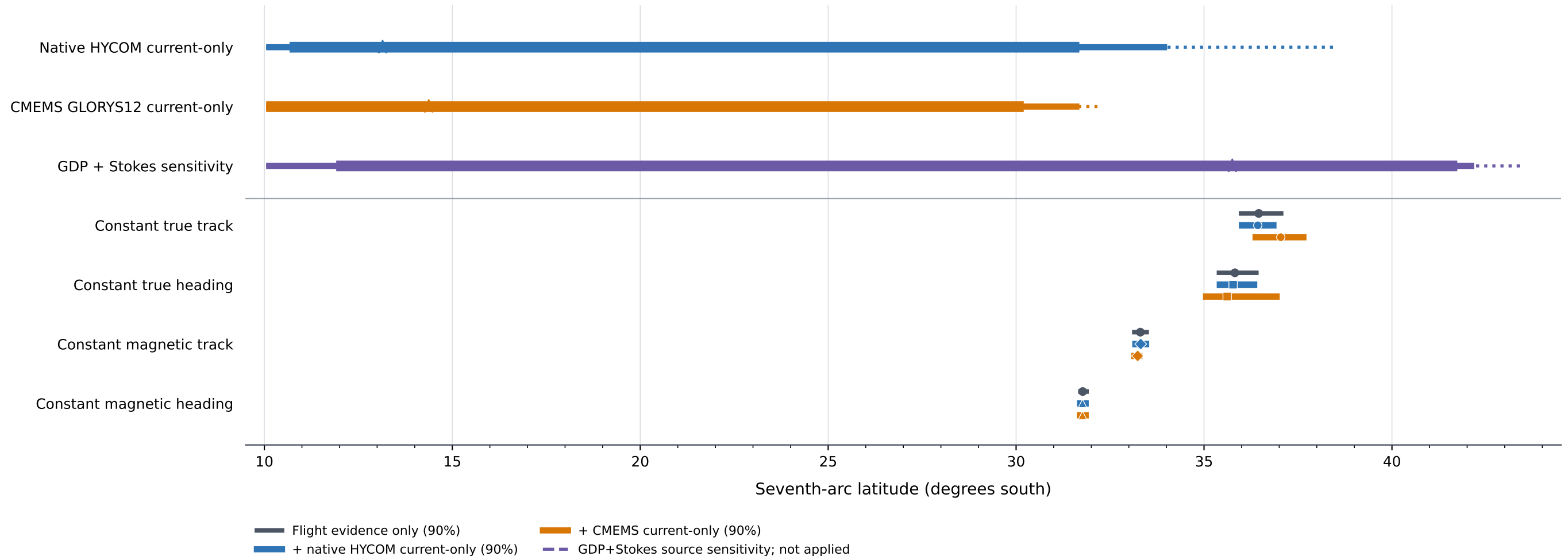
A Source-area probability mass



B Source modes and flight-conditioned means



C Source intervals and inferred crash-location intervals

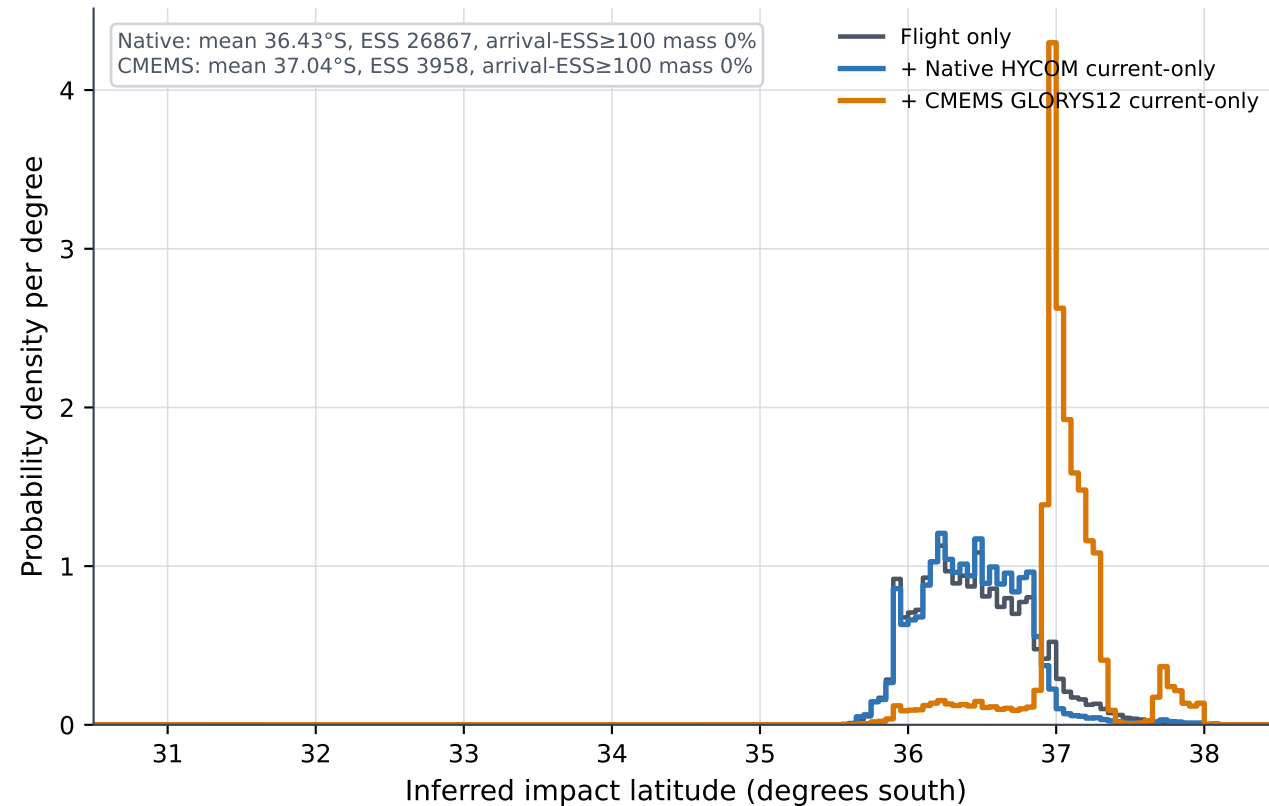


Diagnostic only. A source cell is a candidate crash point only conditional on seventh-arc release at impact.
The flight overlap lies in weakly sampled drift tails, so these curves are not an accepted narrow crash posterior.

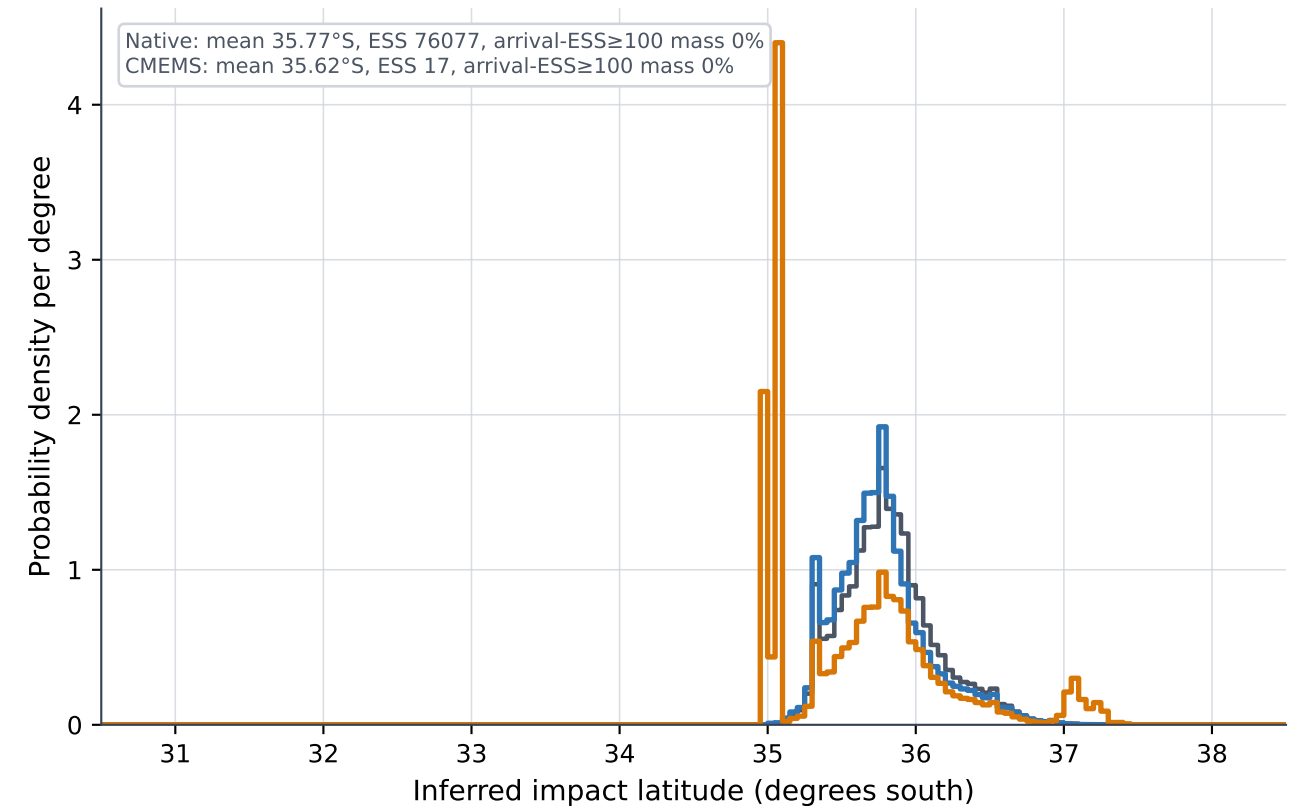
Flight-family-conditioned crash-location densities

Each panel multiplies one converged flight posterior by one source-family likelihood after removing the source prior

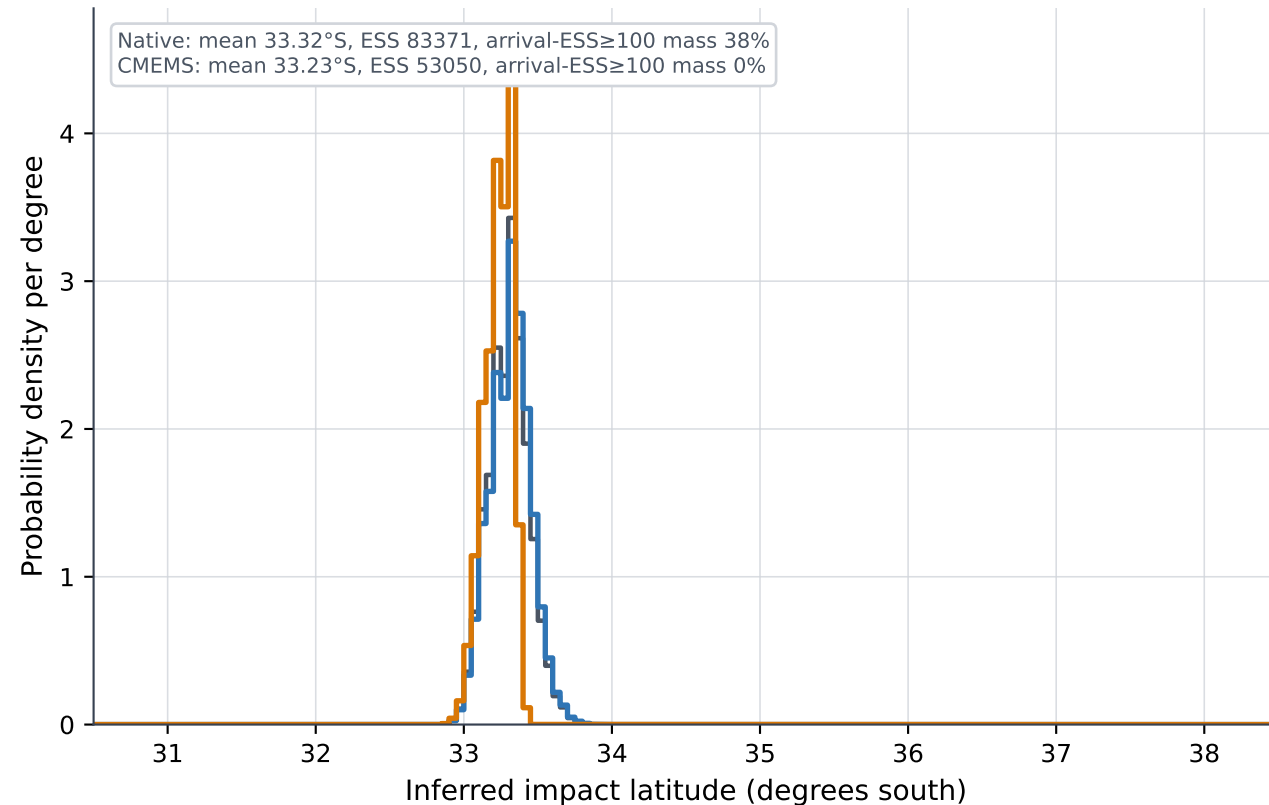
Constant true track



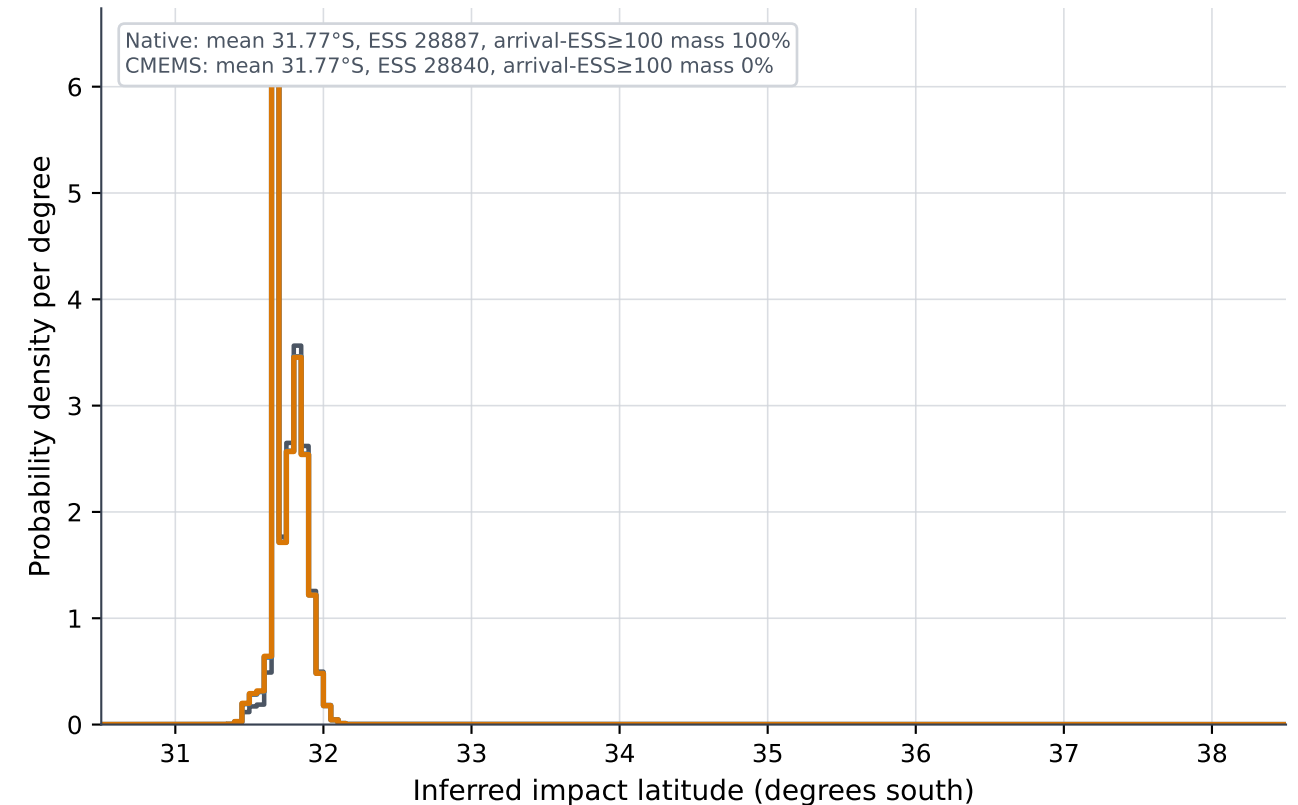
Constant true heading



Constant magnetic track



Constant magnetic heading



The jagged changes are native source-cell likelihoods, not a smoothed fit. Low arrival-ESS mass and probability-floor dependence are retained, not hidden.

Quantitative status and interpretation

The source and crash concepts are related but not interchangeable

A drift source cell is the simulated flaperon's release point on the seventh arc. It is also a candidate crash point only if the flaperon separated at impact at 00:19 UTC. Source-only mass uses debris recovery, the conservative isotope screen, and its explicit arc prior; it does not yet enforce flight feasibility. The inferred crash rows below multiply the converged flight posterior by the source likelihood after removing that source prior. No HYCOM/CMEMS or autopilot-family weight has been invented.

Current family	Mode °S	90% °S	95% °S	99% °S	Terminated	Max arrival ESS	Mass ESS≥200
Native HYCOM current-only	13.15	10.67–31.68	10.05–34.02	10.05–38.46	9.5%	945	91%
CMEMS current-only	14.37	10.05–30.20	10.05–31.68	10.05–32.16	1.1%	494	65%
GDP + coarse Stokes sensitivity	35.75	11.91–41.74	10.05–42.19	10.05–43.48	60.6%	293	12%

Post-hoc family-conditioned inferred crash locations

Flight family	Current	Mean °S / °E	90% latitude °S	Post-hoc ESS	Mass source ESS≥100 / ≥200	Floor-cell mass
Constant true track	Native	36.43 / 89.47	35.93–36.93	26867	0% / 0%	0.0%
Constant true track	CMEMS	37.04 / 88.47	36.29–37.73	3958	0% / 0%	6.8%
Constant true heading	Native	35.77 / 90.64	35.34–36.42	76077	0% / 0%	0.0%
Constant true heading	CMEMS	35.62 / 90.88	34.97–37.01	17	0% / 0%	19.8%
Constant magnetic track	Native	33.32 / 93.98	33.09–33.54	83371	38% / 0%	0.0%
Constant magnetic track	CMEMS	33.23 / 94.05	33.06–33.36	53050	0% / 0%	0.0%
Constant magnetic heading	Native	31.77 / 96.02	31.62–31.93	28887	100% / 7%	0.0%
Constant magnetic heading	CMEMS	31.77 / 96.02	31.62–31.93	28840	0% / 0%	0.0%

Decision: diagnostic only — do not publish a single narrow crash posterior. Native versus CMEMS source TV is 0.392 (overlap coefficient 0.886). The flight families occupy ~32–37°S, while drift-only modes are ~13–14°S. Their intersections rely on the southern drift tails; several CMEMS combinations depend on arrival ESS near one or the recovery probability floor, and one post-hoc family collapses to only tens of effective particles.

GDP remains visible as a separate sensitivity but is not applied: about 61% of its importance-weighted paths terminate at missing field coverage and its replicated shape is unstable.